Sahil Singh

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Summary

Software Engineer specializing in C++, Python, and distributed systems. Experienced in architecting highly concurrent backends, automated testing pipelines, and robust APIs. Built production-ready applications featuring ACID-compliant databases and real-time network pipelines. Passionate about solving ambiguous abstract problems, optimizing algorithmic complexity, and ensuring system reliability at scale.

Projects

OfflineUPIMesh – Distributed Offline Payment Engine *C++20, Crow, ASIO, OpenSSL, SQLite3, CMake*

- Architected a high-performance C++ microservice for mesh-routed offline payments, demonstrating deferred ledger settlement and secure distributed systems concepts across untrusted intermediary nodes.
- Engineered a highly concurrent ingestion pipeline utilizing `std::mutex` locks and `unordered_map` hashing to instantly short-circuit duplicate packet storms (O(1) lookups), efficiently preventing double-spending.
- Implemented TLS-style hybrid cryptography (AES-GCM wrapped in RSA-2048) to guarantee payload authenticity, ensuring ACID-compliant debits/credits via strict SQLite exclusive transactions.

NoJudgement – AI Speaking Practice Platform *Next.js (React), WebRTC, PostgreSQL, Python*

- Developed a scalable web application integrating WebRTC and Socket.io for live peer matchmaking, successfully handling concurrent user sessions and fixing race-condition anomalies on singleton sockets.
- Designed scalable PostgreSQL schemas via Prisma ORM to support automated reporting pipelines, streak tracking, and high-throughput data processing at scale.
- Built real-time STT pipelines integrating Whisper and LLaMA for instant AI-driven feedback, dynamically rendering technical solutions into an intuitive frontend UI.

SantaClaw – Autonomous Execution & Pipeline Agent *TypeScript, Docker, AI SDK*

- Engineered an autonomous execution framework featuring a structured pipeline (intent → repo analysis → planning → execution → diff generation → approval gate).
- Designed a modular testing and staging ecosystem where every automated file modification is isolated and requires explicit human-in-the-loop validation before merging.
- Containerized microservices and shipped to Docker Hub for remote execution, enabling zero-install automated deployments via a single CLI command.

Experience

Software Developer Intern May 2026 – Present
IFFCO (Indian Farmers Fertiliser Cooperative) *Phulpur, Uttar Pradesh*

- Contributing to enterprise-scale internal software systems, leveraging modern full-stack workflows and relational database architectures to support critical business operations.
- Engineered backend APIs and automated testing workflows, designing modular architecture to improve system monitoring and overall operational efficiency.
- Collaborated cross-functionally using agile methodologies to effectively communicate technical challenges and implement optimized solutions in a fast-paced environment.

Web Developer Feb 2026 – Present
Google Developer Groups KIIT Chapter *Bhubaneswar, Odisha*

- Architected event-driven web applications using React and Node.js for a 500+ member community, coordinating closely with peers to consistently ship features on deadline.
- Integrated RESTful APIs and PostgreSQL/Supabase authentication, optimizing upload and data delivery workflows.

Technical Team Member – Web Developer May 2025 – Present
K-1000 *Bhubaneswar, Odisha*

- Maintained scalable backend architecture supporting full CRUD operations; refactored legacy API routes into modular server actions to decrease latency and enhance maintainability.

Education

KIIT University **Class of 2028** (Aug 2024 – Present)
B.Tech – Computer Science (AI & ML) *Bhubaneswar, Odisha*

Technical Skills

Languages	C/C++, Python, Java, JavaScript, TypeScript, SQL
Web / Backend	React, Next.js, Node.js, Express.js, HTML/CSS, WebRTC, REST APIs, Socket.io
Databases	PostgreSQL, SQLite, MySQL, MongoDB, Supabase, Prisma ORM
CS Fundamentals	Object-Oriented Design, Data Structures & Algorithms, Distributed Systems, Concurrency, ACID
Tools / ML	Git, GitHub, Linux, CMake, Docker, PyTorch, Postman, Vercel